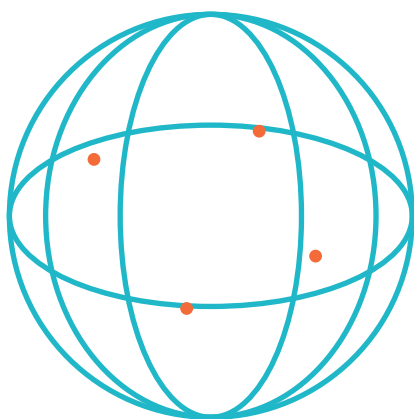


FireAtlas Continuity

An uncertainty-aware MODIS-VIIRS fire activity calendar that turns incompatible satellite records into one transparent, comparable timeline.



HERO GLOBE RETAINED

Core promise

Every calendar value tells the judge what it is: observed, estimated, unknown, or affected by a documented processing gap.

Hero globe: retained exactly as requested, with all existing controls and terrain link preserved.

What changed from the original winning plan

The original plan is strong on provenance and safety. This version strengthens the scientific bridge by harmonizing on a common 1 km grid, uses VIIRS 375 m as a detail layer, adds independent burned-area corroboration, makes the sensor mismatch visible in the UI, and aligns the product directly with NASA's five judging criteria.

Document status: submission-ready blueprint after the data, eligibility, and deployment gates pass. It is a strategy document, not a guarantee of a NASA score.

Primary case studies: Northern California and Punjab-Haryana.

Primary audience: emergency planners, land managers, scientists, educators, and the public.

1. Executive decision

The product should be memorable because the scientific problem is visible, explainable, and useful.

The attached plan already has the right foundation: a working globe, an authentic NASA data archive, a calendar, a provenance ledger, and explicit limits. The main improvement is to make the core contribution easier to see. The judge should understand, within the first minute, that MODIS and VIIRS are not directly comparable because their spatial resolution, sampling, and detection behavior differ. FireAtlas solves that problem with a common-grid sensor bridge and keeps every uncertainty visible.

Final one-sentence pitch: FireAtlas makes the MODIS-to-VIIRS handoff understandable by converting both sensors into one evidence-linked fire-activity calendar, while showing exactly which values are observed, estimated, or unknown.

Keep from the original plan	Change or strengthen	Why it matters
Hero globe and all controls	Add only a small sensor-story overlay	Preserves the project identity and visual impact
Two-region calendar	Use a common 1 km grid before calibration	Makes MODIS and VIIRS spatially comparable
Hatched data gaps	Add observed / estimated / unknown / zero-export states	Prevents the UI from implying that missing rows mean no fire
Ratio and held-out tests	Add matched-grid and baseline comparison	Improves scientific validity
Static site and source ledger	Add a shareable evidence card	Makes results useful in a briefing or review
Safety sentence	Add a separate rapid pulse mode	Supports early-warning relevance without mixing NRT with historical science

Critical correction

Do not compare raw MODIS 1 km counts directly with raw VIIRS 375 m counts. Aggregate VIIRS to a common grid, preserve the 375 m detail separately, and call the main measure VIIRS-equivalent activity.

2. NASA judging alignment

The PDF's internal 100-point tracker remains useful for engineering QA, but the submission should be designed around NASA's published judging criteria.

NASA evaluates projects using Impact, Creativity, Validity, Relevance, and Presentation. Recent global award frameworks also include Best Use of Science, Best Use of Data, Best Use of Technology, Galactic Impact, Local Impact, and Best Use of Storytelling. The exact 2026 award categories should be checked again when NASA publishes its final guide.

Criterion	Judge must see	Evidence to prepare
Impact	A practical way to inspect unusual fire activity and critical periods across two contrasting regions	Two persona walkthroughs; downloadable evidence card; rapid pulse separated from the science calendar
Creativity	A visible sensor bridge instead of an unexplained black-box correction	Common-grid diagram; MODIS-only / VIIRS-only / both / gap mismatch view
Validity	A scientifically defensible calibration with transparent limits	Standard archive metadata; held-out years; baseline comparison; interval coverage; independent corroboration
Relevance	Direct response to the requested burning-activity calendar	Verdict, heatmap, critical dates, raw rows, official links, and selected-area context
Presentation	A clear story that can be understood without reading the method page	Globe-to-calendar demo; low-text UI; captions; 30-second and 90-second cuts

Judge-facing message

A red hotspot is not automatically a comparable measurement. FireAtlas shows how the two sensors disagree, why they disagree, and how the final calendar preserves that uncertainty.

3. Product promise and user journey

The interface should serve three audiences without creating three separate applications.

- Emergency planner: identifies a critical inspection window, checks data completeness, and opens the NASA evidence behind one day.
- Land manager: compares seasonal activity across years and sees where an apparent increase may reflect sensor differences.
- Scientist or educator: downloads the harmonized series, reads the method, and inspects the source rows.

1. See	2. Compare	3. Verify	4. Share
Hero globe shows the living Earth and existing controls.	Region switcher and Sensor Bridge show the MODIS-VIIRS mismatch.	Verdict, heatmap, critical dates, and day drawer reveal the evidence state.	Evidence card carries value, uncertainty, sources, version, and hash.



The 375 m VIIRS layer remains visible as a detail channel because it is valuable for smaller fires and fire structure. The calendar itself uses common-grid units so that the comparison is meaningful.

4. Science and data architecture

The core model must be simple enough to explain and strong enough to test.

Primary harmonized unit: VIIRS-equivalent active-fire cell-days on a common 1 km grid.
Secondary context: raw Fire Radiative Power in MW/day, shown separately.
Detail layer: original VIIRS 375 m detections and raw source rows.

Product choice: use standard MODIS Collection 6.1 and standard VIIRS archive data for historical analysis. Where available, VIIRS 750 m is an especially strong continuity bridge because NASA describes it as a MODIS-algorithm continuity product. If the 750 m archive is unavailable, aggregate 375 m VIIRS into the common 1 km grid and label the transformation explicitly.

Step	Implementation rule	Visible output
1. Import	Keep request metadata, version, date range, bounding box, retrieval date, SHA-256, and row count	Data Sources ledger
2. Filter	Use UTC day; retain type 0 or missing as vegetation-fire candidates; count excluded types separately	Filter summary
3. Deduplicate	Deduplicate within each sensor and day before aggregation	Raw count and unique cell count
4. Common grid	Aggregate VIIRS 375 m to a common 1 km grid; retain raw 375 m detail	Common-grid activity
5. Calibrate	Estimate seasonal or annual VIIRS-to-MODIS relationship only from complete overlap periods	Calibration card
6. Validate	Use leave-one-year-out testing against no correction, single ratio, and seasonal ratio	Method chart
7. Classify	Observed, scaled, unknown, documented processing gap, or complete zero export	Heatmap pattern and badge

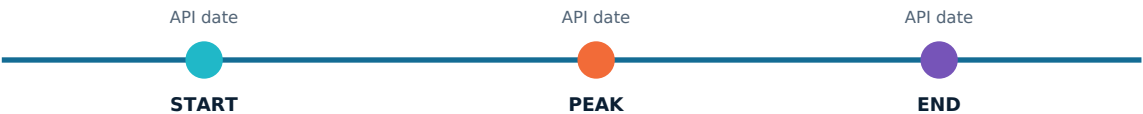
Never combine two observations into one count
When both sensors report the same day, the calendar should use the selected reference observation and show the other sensor for comparison. Summing both would double-count activity.

5. Harmonization and validation method

Every model choice must be visible, reproducible, and allowed to fail when the archive is weak.

Reference logic: use the complete VIIRS common-grid value when available. For a date before the VIIRS archive begins or inside a documented processing gap, calculate a scaled estimate from MODIS and mark it as estimated. For an incomplete export outside a documented gap, return unknown. A complete export with zero eligible rows is an observed zero-detection record, while observation opportunity remains unknown.

Model candidates: monthly ratio, single annual ratio, robust matched-grid ratio, and no harmonization. Select the model only when leave-one-year-out error improves against the baselines. Report median absolute log error, annual absolute percentage error, and interval coverage.



If missing days exceed the threshold, the timeline is withheld.

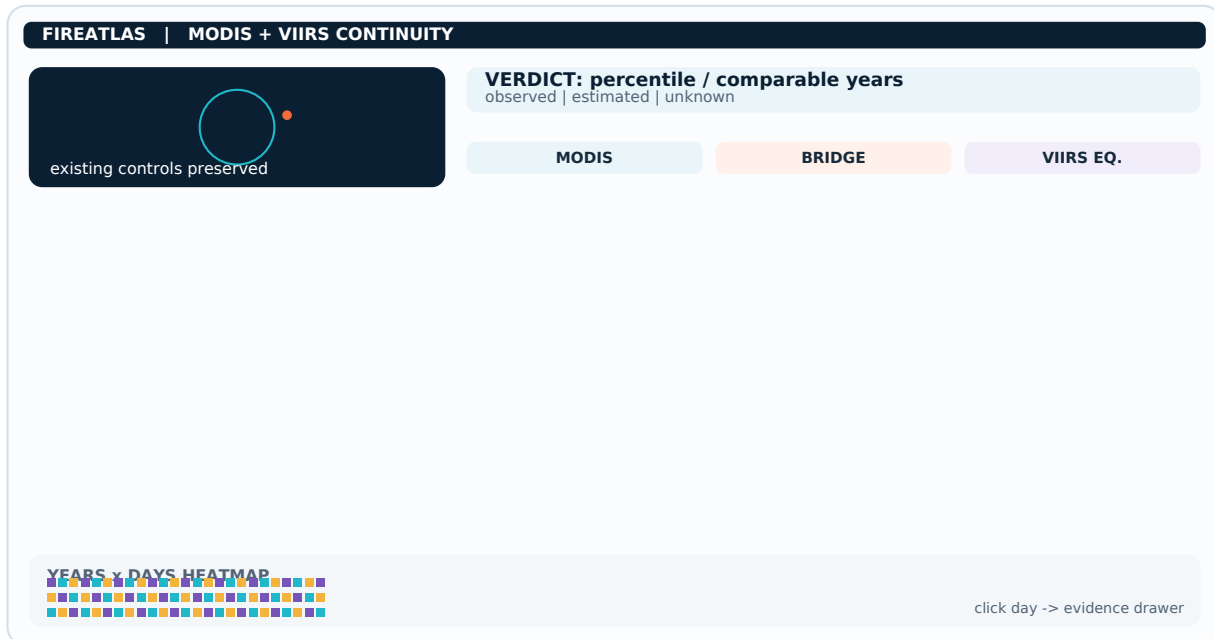
Validation gate	Pass condition	If it fails
Archive completeness	All claimed baseline months have authentic request metadata	Restrict the verdict to verified years and label older history partial
Common-grid agreement	Daily and monthly aggregates reproduce source rows after deduplication	Stop the model claim and show raw sensor comparison
Held-out improvement	Selected method beats no harmonization and single ratio on held-out years	Use the simpler model or publish a comparison-only result
Interval behavior	Coverage is reported and not presented as certainty	Show “calibration weak” beside the estimate
Independent corroboration	MCD64A1 or an official incident record provides separate supporting context	Keep the overlay out and state that active fire is the only evidence

Independent product

Use MCD64A1 Collection 6.1 burned-area data as a lagged corroboration layer only. It should never be treated as a direct ground truth for an active-fire day.

6. UI specification

The page should communicate the result visually before it asks the user to read the method.



Order	Panel	Required content
1	Header	FireAtlas brand, challenge name, region switcher, Method, Sources, data status
2	Hero globe	Existing globe and controls unchanged; small sensor-story overlay only
3	Region and time	Northern California / Punjab-Haryana; month and year; Calendar / Compare / Recent Pulse
4	Verdict	Status chip, percentile sentence, comparable-year count, estimated days, unknown days, official links
5	Sensor Bridge	MODIS raw, common-grid transformation, VIIRS-equivalent result, uncertainty range
6	Heatmap	Years by days; pattern legend; keyboard-readable tooltip; click-to-open evidence
7	Critical dates	Season start, 15-day peak, season end; unavailable state when coverage is insufficient
8	Evidence drawer	Sensor values, source notice, quality state, original NASA rows, download button
9	Footer	Limits sentence, NASA acknowledgement, data version, last build date

7. Visual clarity, accessibility, and mobile behavior

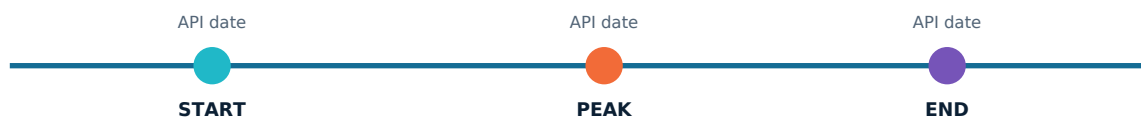
The interface should feel like a scientific instrument, not a report pasted into a webpage.

Color and pattern system: observed = cyan solid; estimated = amber hatch; unknown = gray crosshatch; documented gap = purple hatch; unusual high = orange-red; typical = blue-gray. Every state must also have a label or pattern so color is never the only signal.

Low-text rule: every primary card contains one headline, one value, one sentence, and one action. Technical explanations open in drawers or tooltips. The method page uses diagrams and comparison bars before prose.

Accessibility: keyboard navigation, focus states, ARIA labels for the heatmap, reduced-motion support, high-contrast text, descriptive download names, and a screen-reader summary of each selected day.

Mobile: the globe remains first but compresses to roughly half the viewport height. Controls become a bottom sheet. The heatmap scrolls horizontally. Verdict and quality badges remain visible before the user reaches the calendar.



If missing days exceed the threshold, the timeline is withheld.

The critical-dates timeline should be a visual summary, while the day drawer remains the place for evidence and source rows.

8. Build order and scope control

Finish the evidence path before adding ambitious features.

Priority	Build	Acceptance gate
P0 - essential	Recover missing standard archive windows; preserve provenance; common-grid pipeline; calendar API; verdict; heatmap; day drawer; static public site	Both regions render authentic values; no fabricated rows; no localhost dependency
P1 - highest return	Sensor Bridge; FRP context; uncertainty states; MCD64A1 validation context; shareable evidence card; external usability test	A judge can understand and verify one result in under two minutes
P2 - only after P0/P1	FIRMS Recent Pulse; NOAA-20 supplemental layer; custom AOI; multilingual labels	Separate data mode, separate calibration, documented limits

Do not spend the hackathon on: fire-spread simulation, wind or evacuation recommendations, drone routing, commander dashboards, causal classification, chat interpretation, or a second globe. These features distract from the challenge's core harmonization requirement and create unsupported operational claims.

Eligibility gate

The attached plan correctly asks for a written Local Lead ruling about pre-event repository reuse. Preserve that gate, cite the current 2026 terms, and do not present pre-event work as hackathon work until the local interpretation is documented.

9. Demo and storytelling plan

The story should move from confusion to evidence to practical use.

Time	Visual	Narration
0-5 s	Hero globe	“NASA satellites have watched fire activity for decades, but the record changes when the sensor changes.”
5-15 s	Sensor Bridge	“VIIRS sees smaller and more nighttime fires than the coarser MODIS record, so raw counts are not directly comparable.”
15-35 s	Northern California calendar	Read the live verdict exactly as shown; open one estimated or gap-marked day.
35-50 s	Evidence drawer	Show MODIS value, VIIRS value, quality state, NASA source notice, and original rows.
50-65 s	Punjab-Haryana calendar	Show a different fire regime and explain that hotspots do not establish cause.
65-80 s	Method page	Show held-out error, baseline comparison, and uncertainty coverage.
80-90 s	Evidence card and URL	“FireAtlas makes sensor disagreement visible and keeps every claim traceable.”

Submission package: working public URL; public repository; 30-second captioned video; 90-second narrated demo; 150-word project summary; README; source ledger; data licence statement; AI-use disclosure; and one screenshot each of the globe, verdict, heatmap, evidence drawer, and method comparison.

Presentation rule

Never begin with the method page. Begin with the globe, move to the verdict, reveal the evidence, then show the method that makes the verdict trustworthy.

10. Final judge-evidence matrix

A category should be claimed only after its evidence is visible on the public site or in the submitted repository.

Claim	Public evidence	Internal check
Impact	Two region case studies, persona actions, exportable evidence card, recent pulse separation	Three task walkthroughs and one outside-user test round
Creativity	Sensor Bridge, mismatch categories, uncertainty hatching, 375 m detail channel	Screenshot and short explanation of the bridge
Validity	Common-grid method, archive hashes, held-out comparison, MCD64A1 context, explicit limits	Full test suite; data audit; calibration report
Relevance	Burning activity calendar, unusual-period verdict, critical dates, official links	Challenge terms mapped to visible panels
Presentation	Hero globe, working controls, low-text hierarchy, captioned demo	Incognito deployment rehearsal and mobile check
NASA data use	MODIS C6.1, VIIRS archive, FIRMS acknowledgement, optional MCD64A1	Source URL, version, retrieval date, licence, SHA-256

Final submission sentence

FireAtlas is a research and learning tool that harmonizes MODIS and VIIRS active-fire records into an evidence-linked burning-activity calendar. It helps users inspect historical patterns, unusual conditions, and critical periods while preserving data gaps and measurement uncertainty.

11. References and source checklist

Use human-readable references in the project repository and on the Sources page.

- [1] NASA Space Apps Challenge 2026, "Harmonization of MODIS and VIIRS Hot Spots," official challenge page: <https://www.spaceappschallenge.org/2026/challenges/harmonization-of-modis-and-viirs-hot-spots/>
- [2] NASA Space Apps Challenge, Judging and Awards Guide: <https://www.spaceappschallenge.org/resources/judging-awards-guide/>
- [3] NASA LANCE FIRMS, Active Fire Data and Archive Download: https://firms.modaps.eosdis.nasa.gov/active_fire/ and <https://firms.modaps.eosdis.nasa.gov/download/>
- [4] NASA VIIRS Land Products, NASA Active Fire Product: <https://viirsland.gsfc.nasa.gov/Products/NASA/FireESDR.html>
- [5] NASA Earthdata Forum, FIRMS FAQ: Questions about VIIRS active fire data: <https://forum.earthdata.nasa.gov/viewtopic.php?t=5189>
- [6] NASA FIRMS Fire Alert Notes: <https://firms.modaps.eosdis.nasa.gov/alerts/notes.php>
- [7] Li, F. et al. (2018). Comparison of Fire Radiative Power Estimates From VIIRS and MODIS Observations. Journal of Geophysical Research: Atmospheres. <https://doi.org/10.1029/2017JD027823>
- [8] Fu, Y. et al. (2020). Fire Detection and Fire Radiative Power in Forests and Low-Biomass Lands in Northeast Asia: MODIS versus VIIRS Fire Products. Remote Sensing, 12(18), 2870. <https://doi.org/10.3390/rs12182870>
- [9] NASA Open Data Portal, MODIS/Terra+Aqua MCD64A1 Collection 6.1 Burned Area: <https://doi.org/10.5067/MODIS/MCD64A1.061>
- [10] 2026 NASA Space Apps Participant Terms and Conditions: <https://www.spaceappschallenge.org/legal/>

NASA acknowledgement

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Document control: FireAtlas Continuity | Updated perfected plan | 30 September 2026 | Keep the hero globe feature as-is.